

Congenital Adrenal Hyperplasia



Section I: Scenario Demographics

Scenario Title:	Congenital Adrenal Hyperplasia with Adrenal Crisis		
Date of Development:	02/03/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups		

Section II: Scenario Developers

Scenario Developer(s):	Dr. Quang Ngo
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a rare pediatric presentation that requires important critical care steps in its management
CRM Objectives:	1) Communicate clearly with team members during complicated resuscitation requiring specific, weight-based drug doses. 2) Lead team members effectively through management of critically ill neonate.
Medical Objectives:	1) Recognize hypoglycemia as an important cause or consequence of a toxic neonate. 2) Demonstrate appropriate treatment of hypoglycemia in a neonate. 3) Recognize the electrolyte abnormalities associated with an adrenal crisis. 4) Initiate appropriate steroid and electrolyte corrective therapy for an adrenal crisis.

Case Summary: Brief Summary of Case Progression and Major Events

A lethargic 1 week old presents from home after recurrent emesis and progressive sleepiness. He is hypovolemic, hypothermic, and hypoglycemic. If his hypoglycemia is not quickly corrected, he begins to seize and will continue to do so until the team gives glucose. If they do not, the patient will go on to have a VF arrest. If the team identifies and treats the hypoglycemia, orders blood work, and fluid resuscitates the child, they receive blood results demonstrating hyperkalemia and hyponatremia. If they correctly identify and treat the patient as a possible adrenal crisis, the neonate is safely transferred to the PICU. If they fail to treat the hyperkalemia or fail to administer steroids, the patient will have a VF arrest.

References

- Claudius I, Fluharty C, Boles R. The emergency department approach to newborn and childhood metabolic crisis. *Emerg Med Clin North Am.* 2005 Aug;23(3):843-83.
- Sharma A, Levy D. (2013). Thyroid and adrenal disorders. In J. Marx, R. Hockberger & R. Walls (Eds.), *Rosen's emergency medicine - concepts and clinical practice*. pp. (1689-92). Philadelphia, PA:Saunders.
- Fleisher, G. R. and Ludwig, S. (2010). *Textbook of pediatric emergency medicine*. Retrieved from <http://catalogue.mcmaster.ca.libaccess.lib.mcmaster.ca/>



Congenital Adrenal Hyperplasia



Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 1 week old neonate is brought to the emergency department because his parents are worried that he's been vomiting and not keeping his feeds down. After he vomited his last feed, his parents noted he was quite lethargic and felt cold. His mom states he's been increasingly sleepy since discharge and she's been needing to wake him to feed. In between feeding, he sleeps and doesn't "act like my other 2 kids did at that age." The team is called to assess this patient urgently after being triaged because the nurse felt the patient looked unwell.

B. Scenario Cast & Realism

Patient:	☒ Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates Brief Description of Role

Mother	Answers pertinent questions about child, hovers at the bedside. Cues team to seizures if necessary.
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C. Required Monitors

☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

D. Required Equipment

☒ Gloves	☐ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☐ Venturi Mask	☐ Tube Thoracostomy Kit
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

E. Moulage

None required.

F. Approximate Timing

Set-Up:	5min	Scenario:	15 min	Debriefing:	20 min
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Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Andrew Reynolds		Age: 1 week		Weight: 3 kg	
Gender: ☒ M ☐ F		Code Status: Full			
Chief Complaint: Vomiting feeds					
History of Presenting Illness: Vomiting, not keeping feeds down. Getting more sleepy and parents need to wake him to feed. After last vomit this morning, really lethargic and cold.					
Past Medical History:		SVD at 40 ² , no complications		Medications: None.	
		Mom G3P3, healthy pregnancy, GBS negative			
		Discharged home day 2 of life			
		Breastfed.			
Allergies: None.					
Social History: Lives at home with mom, dad, and two older siblings.					
Family History: Nil.					
Review of Systems:		CNS: Increasingly sleepy, needs to be woken for feeds.			
		HEENT: Nil.			
		CVS: Nil.			
		RESP: Nil.			
		GI: Vomiting, not keeping feeds down. Vomiting between feeds also. Not projectile.			
		GU: No wet diapers since yesterday afternoon.			
		MSK: Nil.		INT: Cold today	
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 180/min		BP: unable		RR: 18-24/min	
T: 35°C		Glucose: 1.5 mmol/L		O ₂ SAT: 98 % RA	
General Status: Appears greyish, unwell. Sunken fontanelle.					
CNS:		PERLA, 2-3mm. Flopping and lethargic with little response. Moving limbs symmetrically.			
HEENT:		Nil acute.			
CVS:		Normal HS, no murmurs. Weak peripheral pulses. CRT 4-5 sec.			
RESP:		Poor respiratory effort. Grunting.			
ABDO:		Soft, non-tender. No hepatosplenomegaly.			
GU:		Nil acute.			
MSK:		No long bone fractures.		SKIN: Greyish.	

Congenital Adrenal Hyperplasia



Section VI: Scenario Progression

Scenario States, Modifiers and Progression			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tach HR: 180/min BP: unable RR: 18-24/min O ₂ SAT: 98 % RA T: 35°C	Floppy, lethargic, making poor respiratory effort. Clinically dehydrated.	<u>Learner Actions</u> ☐ Monitors, warming ☐ IV access, 20 cc/kg fluid boluses ☐ Administer O ₂ ☐ Order septic work-up ☐ Check sugar, give glucose (If D10, give 5-10cc/kg, if D25, give 2-4cc/kg) ☐ Administer antibiotics (ampicillin + cefotaxime or aminoglycoside)	<u>Modifiers</u> - Give 2x20cc/kg boluses → BP registers 60/40, HR to 160 <u>Triggers</u> - Fluids given, blood work ordered, glucose corrected, or 10 min. → 2. Bloodwork back - Capillary glucose not checked by 5 min. → 3. Seizure - Blood work not ordered by 5 min. → 4. VF arrest
2. Blood work back HR → 160 BP → 60/40 RR → 20 T → 35.3 °C	Clinically less dry. Still looks unwell. CRT now 3 sec.	<u>Learner Actions</u> ☐ ID hyperkalemia, do ECG ☐ Treat hyperkalemia: 1) Calcium gluconate 50mg/kg 2) D25 + 1 unit insulin R 3) NaHCO ₃ 1mEq/kg ☐ ID hyponatremia and treat slowly ☐ ID adrenal crisis as most likely diagnosis, treat with hydrocortisone 2mg/kg	<u>Modifiers</u> - Give hydrocortisone → BP to 70/40 <u>Triggers</u> - Don't recognize adrenal crisis by 10 min. → 4. VF arrest - Don't treat hyperkalemia by 10 min. → 4. VF arrest - Correct potassium and give hydrocortisone → End Case
3. Seizure HR → 180 BP → unable RR → no effort O ₂ SAT → 90% despite FiO ₂ T → 35.5°C	Patient having tonic clonic seizures.	<u>Learner Actions</u> ☐ Check blood glucose ☐ Correct glucose (If D10, give 5-10cc/kg, if D25, give 2-4cc/kg)	<u>Modifiers</u> - Patient will keep seizing until glucose given. <u>Triggers</u> - Glucose given → 2. Bloodwork back - Intubation without glucose check → 4. VF arrest
4. VF arrest Rhythm → VF HR → 280, no pulse BP → unable	Patient pulseless with VF rhythm x3 rounds.	<u>Learner Actions</u> ☐ Good quality CPR ☐ Shock VF (2J/kg 1 st shock, 2 nd shock 4J/kg) ☐ Epinephrine 0.01mg/kg ☐ Amiodarone 5mg/kg ☐ ±Intubation	<u>Triggers</u> - 15 min → End Case

Congenital Adrenal Hyperplasia



Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 120	K: 7.5	Cl: 100	HCO ₃ : 10	BUN: 10	Cr: 35	Glu: 1.5
Ca: 2.1		Mg: 0.8		PO ₄ : 1.1		Albumin: 34
VBG	pH: 7.31	PCO ₂ : 32	PO ₂ : 45	HCO ₃ : 18	Lactate: 3.1	
WBC: 18.5		Hg: 140		Hct: 0.66		Plt: 550

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How do you remember strategies for dosing glucose in children? Do you have any “rules” or memory tricks that you use? 2) What are the laboratory abnormalities associated with congenital adrenal hyperplasia? How are they different in secondary adrenal failure? Why?	
Key Moments	
1) Recognition that the neonate is critically ill.	
2) Identification and treatment of hypoglycemia.	
3) Identification of possible adrenal crisis.	